Emigration - a population-migration mod for Civilization VII

Contents

Emigration

Adds more detailed migration and refugee systems to Civilization VII. When a settlement is starving, unhappy, or under siege, people leave. When a settlement is thriving, they move there instead. Population moves between settlements, changing yields, growth, and Influence as it goes. This happens within and between civilizations. Every move is recorded in the notification log with its cause.

A note on the human reality behind this mod

Migration and displacement are abstracted here into game systems. In reality, many people leave home because of war, persecution, disaster, or hardship. This mod aims to acknowledge those realities rather than trivialize them.

As part of creating *Emigration*, I donated to the International Refugee Assistance Project. While I'm not able to sustain a per-subscriber pledge indefinitely, I intend to mark major milestones with donations of time or money within my means. If you are able, please consider supporting organizations such as UNHCR, the IRC, MSF, IRAP, or local refugee and mutual-aid groups.

Subscriber milestones

- **100 subscribers:** Donated $100 to the International Refugee Assistance Project.
- **50 subscribers:** Donated $50 to Médecins Sans Frontières.
- **25 subscribers:** Donated $25 to the International Rescue Committee.
- **10 subscribers:** Pledged one hour of volunteer time for a refugee-support, humanitarian, or mutual-aid organization.
- **Release donation:** On upload, made an initial donation to the International Refugee Assistance Project.

Anonymized receipts for these donations will be added to the repository.

Mechanics

- **Compatible with 1.4.1.**
- **Migration scoring.** Each settlement gets a score from its yields, happiness, war weariness, and government passives. Population moves from low-scoring settlements to higher-scoring ones each turn.
- **Displacement.** Damage, pillaging, sieges, starvation, unrest, and disasters generate refugees. Destination priority: own civilization, then neutrals, then the attacker.
- **Attrition.** Sustained sieges, famine, war, or disaster kill population that cannot relocate. The death chance accumulates over several turns instead of resolving in one.
- **Delay.** Arriving refugees sit idle before joining the host's working population. They produce no yields but carry a support cost, and appear on the map lens and city readout.
- **Distance penalty.** Move probability falls off with distance, so migration favors nearby settlements over cross-map jumps.
- **Borders and policy.** Open Borders agreements and Pro-/Anti-Immigration policies raise or lower migration rates, settlement chance, retention, and integration cost.
- **Integration cost.** Each incoming migrant applies a temporary happiness and gold cost to the host. A congestion penalty caps how much any single settlement can absorb.
- **Origin tracking.** Settlements record the origin civilization of their population. Migrants integrate over time, diasporas can return home, and a persistent foreign population can form a cultural enclave.
- **Attribution.** Every move stores the factor that decided it (prosperity, distance, safety, open borders, allies, asylum, crisis, or war), surfaced in notifications, city readouts, pressure warnings, crisis reports, and a persistent log.
- **Game speed.** All turn-based pacing scales automatically from Online to Marathon.

Migration dashboard

Available through an optional dock button or the Demographics mod's interface. Tabs cover the Migration Network, Net Migration, Why People Move, Settlements, Immigration Policies, Guide, Notifications log, and Migration notifications.

It runs in the UI VM (GameFace JS) each turn. This is not a UI-only mod: population moves and yield/Influence changes are real gameplay writes (§12).

Documentation

- Migration mechanics overview: ../emigration-docs/DESIGN.md
- Engine behavior checks and verification notes: ../emigration-docs/FINDINGS.md
- In-game validation checklist: ../emigration-docs/testing-requirements.md
- Civ VII modding mechanics & limits: ../emigration-docs/civ7-mechanics-and-feasibility.md
- Leader/civ ability & memento interactions: ../emigration-docs/leader-civ-memento-interactions.md
- Advanced migration formulas: ../emigration-docs/algorithmic-improvements.md
- Interactive systems: ../emigration-docs/interactive-extensions-design.md, ../emigration-docs/interactive-extensions-implementation.md

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System Guide and Feature Reference

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1. What it does (player-facing)

Each turn the mod scores every visible city and moves population from the lowest-scoring settlements to the highest:

- **Peacetime.** Happiness is the largest factor; unhappy or low-yield cities lose people to happier, wealthier ones. No war required.
- **War refugees.** A city taking district damage or with pillaging in its borders sheds people fast, fleeing away from the nearest invader.
- **Concurrent causes.** War, disaster, and economic pressure are scored independently and at the same time (§2), so one city can shed war refugees and economic migrants in the same turn. Each move has one cause.
- **Cross-civilization.** People move between civs, not only within your own.
- **Regional.** Distance-penalized: people move to nearby better settlements, not across the map.
- **Consequential.** Receiving migrants costs the destination happiness and gold while it integrates them, so magnets converge instead of growing without limit. Holding unsettled migrant units also costs.
- **Speed-aware.** Pacing scales with game speed (§2); the game-time rate is the same on Quick, Standard, or Marathon.

Reported in-game (toasts + Demographics graphs) and in the dev log, e.g. `EMIGRATION 1 population point ( 30,000 people) left Rome (Romans) for Carthage (Carthaginians)`.

Quick reference: what counts

Default settings (most tunable, §10). Also on the dashboard's **Guide** tab.

What makes people leave

	Counts?	
Unhappiness / low yields	✓	Main peacetime driver; happiness weighs most, low per-capita yields add to it (1.4.1 also suppresses unhappy cities' yields)
Empire-wide war weariness	✓	A modest empire-wide push, on top of per-city violence
War damage to districts	✓	Read from game state (fog-independent); more damage, more push
Being besieged or attacked	✓	Only the besieged city; fires when any of its districts is besieged or overrun
Attacked by a city-state / Independent Power	✓	Same conflict pressure as a major-civ war
Pillaged tiles in the city's borders	✓	Damaged improvements on the city's own plots (polled, fog-independent)
Starvation	✓	Negative net food: most flee, some die, until food recovers
Plague / disease	✓	An infected city loses people; migrants can carry plague onward
Natural disasters (floods, volcanoes)	✓	A capped per-city penalty; a strike hits every city around its epicenter
Overcrowding in a tall city	✓	Urban population over a threshold; softened by per-leader tuning

What attracts

	Attracts?	
Higher prosperity (food, production, gold, science, culture)	✓	Per-capita weighted yields above nearby cities
Higher happiness	✓	Weighted most; in the shaped model it saturates, so no runaway magnet
Pro-Immigration stance	✓	Raises inbound pull, earns Influence
Open Borders agreement	✓	Cross-civ pull bonus
Being nearby	✓	Distance-penalized

Who participates

	Participates?	
Your civilization	✓	Sends and receives like any major civ
Towns	✓	Same as cities
Your own cities (internal migration)	✓	Coloured separately on the dashboard
Other major civilizations	✓	All simulated from turn one, met or not
City-states / minor civs / Independent Powers	✗	Don't send or receive (but attacking a city still drives its people out)
Unmet civilizations	✓	Simulated, masked in the UI by default

Behavior

Migration between civilizations	✓	Throttled by borders, distance, and stance
Distant AI-vs-AI wars	✓	Only when they damage, besiege, or pillage a city's own territory
Fighting outside a city's borders	✗	Never drives its emigration; war pressure is territory-scoped (§3)
Anti-Immigration stance retains people	✓	More retention + Production, less Influence
Closed Borders reduces cross-civ flow	✓	Far fewer cross without an Open Borders agreement
Population & yields actually change	✓	Real per-turn gameplay writes
Pacing adapts to game speed	✓	Cooldowns, ramps, transit, thresholds scale (§2)
Any layer tunable or off	✓	Presets + ~57 knobs, Options Mods Emigration
Migrants arrive instantly	✗	They travel; arrival lags with distance
Absorbing migrants is free	✗	A temporary, decaying happiness + gold cost
War can empty a city to zero	✗	Displacement is capped (`siegeLossCapPct`) above the rural floor; only a capture empties/transfers. Crisis deaths (separate, unfloored) can wear rural pop down, but the urban core survives

Identity, integration & return

Settlements remember origins	✓	Running composition by origin civ; a per-tile mosaic on the Ethnic Composition lens (Shift+E); kept through capture
Newcomers integrate over time	✓	Each non-owner origin drifts toward the owner per turn (Options ethnic integration)
War / unrest keeps a community distinct	✓	Integration stalls at war with the homeland, slows in unrest
Diasporas return home	✓	When the homeland is at peace and prospering, a fraction return, moving real population (Options return migration)
Refugee waves prompt a decision	✓	A rare modal (welcome / settle the frontier / turn away), capped per age, dismissible
Migrations written as history	✓	The Migration Chronicle tab; unmet civs framed as hearsay

Scope & limits

Changes AI or replaces base-game files	✗	Additive only
Moves population instantly	✗	Distance-penalized
Lets you place individual migrants	✗	Simulated; you shape flows with yields and stances

Lets one civ snowball the map	✗	Three brakes: field-relative scoring, a congestion headwind, and an anti-snowball headwind on cross-civ inflow (all tunable)

FAQ

- **Where do people go?** The nearest higher-prosperity settlement they can reach.
- **Where do war refugees flee?** Away from the nearest enemy: own civ, then neutrals, attacker last.
- **How many, how often?** War/disaster refugees flee every turn; voluntary migration is gradual. Each civ has its own per-turn budget, so simultaneous wars don't throttle one another.
- **Capturing or losing a city?** It keeps its origin mix; only a capture transfers it.
- **Why the sudden drop?** A toast names the cause; the per-city readout shows the pressure mix.

Post-war recovery

- **Will a war-shrunk city grow back?** Yes. Displacement moves population points, never razes districts or buildings (only conquest does). It regrows via food growth and immigration once fighting stops.
- **Do the same refugees return?** No repatriation; it regrows from new residents.
- **Does repairing pillaged tiles restore population?** No. Repair removes the pressure (faster recovery) but adds no population back.
- **How far can war shrink a city?** Displacement is capped at `siegeLossCapPct` (60% by default) of onset population. Crisis deaths are separate and uncapped and can wear rural pop past that, but only a capture takes the city. (The cap is a fraction, speed-invariant.)
- **Fastest recovery?** Make peace (violence decays in ~2–3 turns), repair pillaged tiles, raise happiness.

Migration in transit

Moves aren't instant: transit lag (`transitLagTurns`, distance- and speed-scaled) leaves a migrant between cities.

- **Lifecycle.** Departure removes the source's rural point immediately (losing that tile's yields). In transit the migrant belongs to no city (no yields, no upkeep). The integration cost is paid by the destination on arrival.
- **Per-city caps.** A city loses at most `maxLossPerCityPerTurn` and gains at most `maxGainPerCityPerTurn` per turn (the inbound cap bounds total intake). Both scale with the intensity preset. Deaths aren't counted against either.
- **Arrival isn't guaranteed.** An arrival into a full city waits and retries (longest-waiting first); if the destination is razed/captured or still full after several turns, the migrants perish in transit. The cap is never overrun.
- **How long.** 1–4 turns at Standard (`transitHexPerTurn` 5 hexes/turn), longer on slower speeds; war/disaster refugees take at least 1.
- **What you'll see.** Gross Emigration can tick up before Immigration catches up. Net Migration counts only settled, cross-civ moves.

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2. How it works: the per-turn loop

On every `PlayerTurnActivated`:

1. **Per-civ costs** (`chargePerTurnCosts`) run for whichever civ's turn it is: the decaying integration cost and the migrant-holding penalty (§7).
2. **The emigration pass** (`runPass`) runs once on the local player's turn (gated by `turnInterval`):
    1. Decay accumulated violence and disaster distress (rates game-speed-adjusted, below).
    2. Collect signals: one `CitySignal` per met city (§3).
    3. Rank by Prosperity: score every city, sort descending (§3, §5).
    4. Advance state: a monotonic turn counter (for scaling), prune/tick cooldowns, compute per-owner populations (for congestion).
    5. Process each source as two concurrent tracks (below), each civ bounded by its own per-turn move ceilings (`civMoveCeilings`, a runaway/perf safety net, not the pacing knob): `maxMovesPerTurn` +

`movesPerCity` · (settlements) for the voluntary ceiling, `movesPerSiege` · (cities in crisis) for the crisis ceiling. Ceilings are per-civ, so simultaneous wars don't compete for one global budget.
6. Persist state to `GameConfiguration`; surface feedback (§9).

3. **Events.** Subscribed at boot: `DiplomacyDeclareWar`/`MakePeace` feed the aggressor map (§6a); `RandomEventOccurred` feeds disaster distress and a named alert (§6c).

Two concurrent tracks: voluntary vs crisis (`splitTracksEnabled`)

Each source is evaluated as two independent systems every pass, so both can fire in the same turn toward the same destination:

- **Crisis** (war / disaster): flees every turn (no bar, no cooldown), bounded by `warSurgeMax` and the cumulative `siegeLossCapPct`. Cause is *disaster* when disaster distress dominates, else *war*.
- **Voluntary** (prosperity / unhappiness): accumulates pressure toward `emigrationBar`, moves one point on crossing it, then rests for `cooldownTurns`. Cause is *unhappiness* when happiness is low, else *prosperity*.

Each track has its own per-civ budget (`splitBudgetsEnabled`). Every record still carries one cause (concurrency = multiple records), so by-cause telemetry is unchanged. The city readout shows the live mix ("War 60% · Prosperity 40%", `splitUiReadoutEnabled`). All three flags default on; off restores single-cause-per-pass.

When a source has no viable destination (the outlet, §6d), a sufficiently distressed source builds attrition pressure and eventually loses a rural point with no destination (a death, not a move).

Game speed (all turn-based pacing scales, `gameSpeedTuningEnabled`)

The engine paces in turns, but game speed stretches the same game-progress over a ~6× range of turn counts (`GameSpeeds.CostMultiplier`). Uncorrected, the mod would be calibrated for Standard and drift elsewhere. Pacing is scaled by the speed scalar **S** so migration feels the same in game-time at any speed:

Speed	CostMultiplier	S	cooldown 8 →	bar 30 →
Online	50	0.5	4	15
Quick	67	0.67	5	20
Standard	**100**	**1.0**	**8**	**30**
Epic	150	1.5	12	45
Marathon	300	3.0	24	90

- **Turn-count durations ×S:** `cooldownTurns`, `siegeRampTurns`, `transitLagTurns`.
- **Pressure thresholds ×S:** `emigrationBar`, `attritionThreshold`.
- **Decay re-based to `d^(1/S)`:** `violenceDecay`, `disasterDecay`.
- **Never scaled:** `siegeLossCapPct` and intensity thresholds (a siege costs the same fraction at any speed), yield weights, friction, and the per-turn move ceilings.

It is automatic: reads the active speed once via `Configuration.getGame().gameSpeedType` → `GameInfo.GameSpeeds.lookup(...).CostMultiplier`, caches it, fail-safe to S = 1 if unreadable. Gated on `gameSpeedTuningEnabled` for rollback. See `emigration-game-speed.js`. A separate, default-off `gameSpeedScalePopulation` flag normalizes the §4 people-scaling exponent (cosmetic, cross-mod; see §4).

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3. The signals & the Prosperity score

`emigration-cities.js` builds a `CitySignal` per city (owner, population, rural pool, urban population, per-capita yields, net happiness, unrest, starvation, siege, war, accumulated violence, accumulated disaster distress, infected flag). `emigration-prosperity.js` turns it into a score. The default (legacy linear) model:

$$
\begin{aligned}
P &= (Q + h\,\lambda_h - n\,\lambda_n)\left(1 + \frac{s}{100}\right),\\
Q &= \frac{f\,w_F + p\,w_P + g\,w_G + sc\,w_S + c\,w_C}{n},\\
s &= v + d + \sigma + \tau + u.
\end{aligned}
$$

P is prosperity, Q per-capita productiveness, h net happiness, n population, s the summed situational percentage from violence, disaster, siege, starvation, and unrest.

Higher = more attractive. Happiness dominates (weight `localHappinessFactor`, default 6). The situational multiplier is where war/violence, disasters, sieges, starvation, and unrest bite. §5 replaces the happiness term and the violence penalty with more nuanced versions when their flags are on. The magnitude of the negative situational percent is exposed as `distress(s)`, which drives the outlet (§6d) and weights the readout's cause mix.

Polity signals (`emigration-polity.js`): happiness stages, governments, celebrations (1.4.1)

1.4.1 reworked happiness, governments, and celebrations, so the model reads three more signals (bounded, additive, behind `polityModelEnabled`; set false for exact pre-1.4.1 scoring):

- **Happiness stage:** each settlement's 5-stage ordinal (Angry −2 … Ecstatic +2), bucketed against `GameInfo.HappinessStages` like the base-game banner. Adds a magnitude-insensitive pull/push (`happinessStageWeight`) so 1.4.1's sharper swings register without re-tuning the raw-happiness knobs.
- **Celebration:** a civ in a Golden Age is a stronger attractor (`celebrationPull`), from `player.Happiness.isInGoldenAge`.
- **Government:** a small clamped per-government lean (`governmentWeight`, `governmentLeanCap`) that breaks ties; most of a government's effect already reaches the model through happiness/yields.
- **War weariness:** a war-weary civ's settlements take a modest empire-wide push (`warWearinessModifier`), distinct from and dominated by the in-border violence terms.

Read once per civ per pass and denormalized onto each `CitySignal` (`stage`, `polity`).

Violence (`emigration-violence.js`): polled, fog-independent

War-driven emigration keys on actual violence inside a city's borders, not on the empire being at war, and is symmetric for player-watched and distant AI-vs-AI wars because it reads game state, not visibility-gated events:

- **City under attack:** polls the center district's health (`getDistrictHealth/…MaxHealth`), readable for all players regardless of line of sight. Fresh damage spikes (`vwAssault`); standing damage sustains a siege (`vwSiege`).
- **Pillage:** damaged improvements on `getPurchasedPlots()` add standing pressure (`vwPillage`).
- The score accumulates and decays (`violenceDecay`, adjusted to `d^(1/S)`): a sustained siege builds, a lone raid fades in ~2–3 turns of game-time. With Algorithm D on, the curve also escalates with siege duration (over `siegeRampTurns`, ×S) and is capped in total (§5-D).

Territory-scoped: all three signals read only the city's own footprint (`districtDamageFrac` / `districtBesieged` match by `owner:id`; `pillagedCount` scans the city's own `getPurchasedPlots()`). A field battle in neutral land, a war elsewhere, a tile the city doesn't own, or a distant AI-vs-AI war never moves its people. The civ-wide `sig.atWar` flag is not a cause (dev-log label only); `fleeVector` is gated on the city's own violence. One boundary case counts: a district flagged besieged by units just outside its borders, which `siegeBesiegedFloor` keeps gradual.

Geography (`emigration-geography.js`)

- **Distance decay:** `−distanceFactor × hexDistance`.
- **Flee-from-invader:** when violence crosses `violenceFleeThreshold`, refugees prefer destinations away from the nearest enemy (`fleeFactor`).
- **Aggressor preference:** own civ > neutral > attacker, when Feature 1 is on (§6a).
- **Open Borders flow bonus:** a modest cross-civ pull bump between civs with a base-game agreement (`openBordersBonus`, §6b).

The destination decision (`emigration-pull.js`)

Two bounded channels over the prosperity gradient and friction terms:

$$
\begin{aligned}
\mathrm{Pull}(s,d) &= \Big(\Delta\mathrm{Pros}(s,d) + \mathrm{Tilt}(s,d) - \mathrm{Friction}(s,d)\Big)\cdot \Pi(s,d),\\
\Delta\mathrm{Pros}(s,d) &= \mathrm{Pros}(d) - \mathrm{Pros}(s),\\
\mathrm{Tilt}(s,d) &= \mathrm{clamp}(\mathrm{asylumTilt}(s,d), -\mathrm{tiltCap}, \mathrm{tiltCap}),\\
\Pi(s,d) &= \mathrm{clamp}(\mathrm{openness}(d)\cdot \mathrm{retention}(s)\cdot \mathrm{permOpenBorders}^{ob}\cdot \mathrm{permAlly}^{al}\cdot \mathrm{permWar}^{wa},\ \mathrm{permeFloor}, \mathrm{permeCeil}).
\end{aligned}
$$

In permeability Π, openness(d) is the destination's inbound throttle and retention(s) the source's cross-civ outbound throttle, the two halves of the Anti-Immigration stance (§6b). Both are 1 unless border policies are on, and retention applies only cross-civ.

Friction:

$$ \begin{aligned} \mathrm{Friction}(s,d) =&\ \mathrm{baseReluctance}

- \mathrm{perExtraPop}\cdot\max\left(0,\mathrm{pop}(d)-\mathrm{pop}(s)\right)
- \mathrm{cityStateBarrier}
- \mathrm{poachBlock} \ &-\ \mathrm{geoAdjust}(s,d)
- \mathrm{congestionFor}(d)
- \mathrm{dominanceFor}(d). \end{aligned} $$

dominanceFor(d) is the anti-snowball headwind: 0 unless the destination civ runs ahead of the world-average civ, then $\mathrm{antiSnowballWeight}\cdot\max\Big(0, \frac{\mathrm{pop}_{\mathrm{civ}}(d)}{\overline{\mathrm{pop}}_{\mathrm{civ}}} - \mathrm{antiSnowballThreshold}\Big)^{\mathrm{antiSnowballExponent}}$. It applies only to cross-civ inflow into a runaway leader, never to its own outflow or internal moves. Tunable (Off / gentle / standard / strong + threshold).

War is not a hard gate: a besieged city has low prosperity and a flee vector, then passes through the same pull equation. People can emigrate to any civilization.

The Prosperity map lens (`emigration-prosperity-lens.js`, `emigration-prosperity-tooltip.js`)

A self-registering lens shades the world by prosperity tile by tile: each plot is scored from its own per-plot yields (`GameplayMap.getYields(plotIndex, playerID)`), normalized against the world plot field, and painted in buckets, with a per-city fallback when per-plot yields aren't available. Hovering a plot adds the reading to the tooltip.

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4. Population scaling (Demographics alignment)

`emigration-population.js` converts abstract population points into representative people using the identical formula to the Demographics mod, grounded in Civ VII's own per-era growth formula (the food cost to grow a settlement, which differs by age):

```
W(N, era)           = Σ cost(1..N) for era's {flat, scalar, exp}   // the game's real per-era growth cost
eraParams(age, pct) = blend(prev-era, this-era params)             // continuous across age boundaries
scaleCityPopulation = POP_K × W(size, eraParams) × megacity × overtime, then soft-capped to the era max
```

Each age uses the game's real growth parameters (Antiquity / Exploration / Modern), so a settlement reads at a sane size for its age with a smooth hand-off at each boundary. A Modern-only megacity term lets the largest cities reach the real 10–38M range; an endgame term keeps figures growing past the natural end; a soft per-era ceiling caps the result. There is no turn-based multiplier, so the figure doesn't drift with game speed. A moved point is reported as the marginal people it represents (`scale(pop) - scale(pop-1)`), and its small per-event variation leans on the source's real happiness and urban/rural mix (its name only as a tie-breaker). `formatPeople` renders "30,000 / 1,300,000 / 240,000,000". `moveRural` performs a relocation; `removeRural` removes a point with no destination (the outlet's death, §6d), using the same rural-population accounting as the game's own starvation shrinkage.

> **Always aligned with Demographics.** Both mods carry the identical scaling, pinned bit-for-bit by a cross-mod parity test (`tests/scaling-demographics-parity.mjs`). Because scaling is keyed to the age,

not the raw turn count, the figure is the same on Online, Standard, and Marathon. The turn-based pacing knobs (§2) are still scaled by game speed and on by default.

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5. The advanced model (algorithms & per-civ tuning)

Four algorithms plus a per-civ tuning table refine the baseline, all on by default (each switchable in Options). Full math + before/after numbers: ../emigration-docs/algorithmic-improvements.md.

A. Shaped happiness (`happinessShaped`)

The linear `happiness × 6` term let pure-happiness sources run away (Franklin's Glass Armonica, +15 happiness/ally, made a ~50× magnet). The shaped model is field-relative (measured vs the world mean), saturating on the pull side and steep on the misery side (`tanh`), and makes happiness amplify the economy (bounded multiplier + `happyFloor`) rather than dwarf it. Franklin drops to ~2× while unhappy cities still shed strongly.

B. Overcrowding discount (`overcrowdDiscount`)

The probe (§12) confirmed population costs zero happiness per head; a tall city's unhappiness is overcrowding past a density threshold, and `getYield` is the net, post-penalty value, so unhappiness double-hits. The discount credits back density-driven unhappiness via `urbanPopulation` vs `overcrowdThreshold`.

C. Congestion headwind + leader variance (`congestWeight`)

A structural anti-runaway brake that can't be out-golded: a civ absorbing many migrants becomes a less attractive further destination, scaling with its per-capita assimilation load. Two leader-variance knobs ride the assimilation cost via the civ table: `integrationSpeed` (load decay) and `assimilationEase` (gold cost).

D. Capped, time-gated war displacement (`warSiege`)

Fog-independent violence made war a bloodless depopulation tool. The siege model tracks siege tenure, escalates the penalty from `siegeFloor` to full over `siegeRampTurns` (×S), and caps total war loss at `siegeLossCapPct` of onset population (the remnant digs in), so a city can lose substantial population but can't be emptied without a capture.

The civ tuning table (`emigration-civ-tuning.js, civTuningEnabled`)

A small, auditable registry of bounded per-leader/per-civ nudges, keyed on the GameInfo leader string (`_ALT` personas normalized; leader overrides civ). Fields: `happinessPull`, `integrationSpeed`, `assimilationEase`, `overcrowdDiscount`, `warRetention`, `sourceBias`. Shipped entries target outliers: Franklin `happinessPull 0.75`, Isabella `0.85+ease 1.2`, Xerxes `ease 1.25`, Khmer `sourceBias 1.5`, Pachacuti `overcrowdDiscount 0.5`, Norman/England `warRetention 1.4`, etc. None can cause a runaway; the structural guarantees live in the algorithms.

Brush & Blade coverage. The table extends to the expansion's civs/leaders, abilities read from the DLC game files and mapped to the same six fields. Conquest economies pay more to absorb spoils (Assyria/Bulgaria/Ottomans/Pirate Republic; Alexander/Genghis Khan/Edward Teach, `assimilationEase` 1.2–1.25), while Bolívar drops to `0.85`. Fortification-defensive civs hold population under siege (Dai Viet/Sengoku `warRetention 1.4`), and Toyotomi (double defensive damage) sheds it (`warRetention 0.85`). Happiness/celebration magnets are damped (Heian/Silla, Himiko `happinessPull 0.85`); tall/few-settlement shapes are shielded from the density penalty (Carthage/Nepal/Qajar); high-growth Shawnee and FOOD-penalized Napoleon get a small `sourceBias` cushion. Civs/leaders with no migration-relevant outlier (Iceland, Tonga, Great Britain; Ada Lovelace, Gilgamesh, Lakshmibai, Friedrich) stay neutral.

Flatten knob (`civTuningStrength`, default 0.7). A global control that compresses every profile toward neutral: `1.0` = the full table, `0` = fully flat (same as off). It interpolates each field toward its neutral, preserving relative ordering while shrinking the absolute spread. The default `0.7` keeps each civ's character but trims divergence ~30% as extra anti-snowball margin; exposed as a Scope tunable.

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6. Interactive systems (on by default)

6a. Aggressor-aware war refugees (`aggressorPenalty`, 0 = off)

When civ A attacks civ B, B's refugees prefer B's own cities first, then any civ other than A, and treat A as a last resort. The aggressor is read from the public `DiplomacyDeclareWar` event (`actingPlayer` declared on `reactingPlayer`), persisted as a victim→aggressors map in `emigration-war.js` and cleared on peace. The preference (`ownCivRefugeeBonus` toward own civ, `-aggressorPenalty` for the attacker) is folded into `geoAdjust` only for cities under violence.

6b. Immigration-stance policies + Open Borders agreements

Two distinct levers control cross-civ immigration.

Your stance (a policy card, `bordersEnabled`). Slot **Pro-Immigration Stance** or **Anti-Immigration Stance** (renamed from "Open/Closed Borders" to avoid colliding with the base game's Open Borders agreement). A database component (`data/emigration-policies-{antiquity,exploration,modern}.xml`, one file per age) adds the traditions, one per age, available to every civ. They unlock from a mid-age civic node (Antiquity **Citizenship**, Exploration **Economics**, Modern **Social Question**). (Internal trait IDs keep `TRADITION_EMIG_OPEN/CLOSED_BORDERS_*`.) The two stances are asymmetric:

- **Pro-Immigration Stance:** +50% immigration into your cities (`immigrationOpenness(destOwner)`) plus a native +1/+2/+3 Influence `TraditionModifier`.
- **Anti-Immigration Stance:** throttles inbound immigration to 40% (floored at 0.15) and retains your own people (cross-civ outbound pull cut to 60%, `emigrationRetention(srcOwner)`), plus a native +2/+3/+4 Production in every city and a −2/−3/−4 Influence penalty.

The migration % and retention are custom UI-VM mechanics; Influence and Production are native `TraditionModifiers` (`data/emigration-policies-gameeffects.xml`), so they show on the card and in the yields breakdown.

Diplomatic Open Borders (a flow bonus, `openBordersBonus`). When two civs hold an active base-game Open Borders agreement, migration between them is eased both ways (checked in `emigration-geography.js`). Console check: `emigration.openBorders(aPid, bPid)`.

Governments no longer separately affect emigration.

Policy cards by age (shipped) Age-scoped in `data/emigration-policies-{antiquity,exploration,modern}.xml`, native per-turn yields in `data/emigration-policies-gameeffects.xml`. Names below are the in-game display names.

Antiquity cards

Policy (Civic)	Effects (native + migration)
Pro-Immigration Stance(Citizenship)	+1 Influence/turn Migration: pull x 1.5(openBordersOpenness)
Anti-Immigration Stance(Citizenship)	-2 Influence/turn, +2 Production/city Migration: inbound pull x 0.4 (floor 0.15);own cross-civ outbound x 0.6 (retention)

Exploration cards

Policy (Civic)	Effects (native + migration)
Pro-Immigration Stance(Economics)	+2 Influence/turn Migration: pull x 1.5(openBordersOpenness)
Anti-Immigration Stance(Economics)	-3 Influence/turn, +3 Production/city Migration: inbound pull x 0.4 (floor 0.15);own cross-civ outbound x 0.6 (retention)
Talent Attraction(Inspiration)	+1 Science/turn Migration: +1.5 Science poolper arrival
Cultural Magnetism(Society)	+1 Culture/turn Migration: +1.5 Culture poolper arrival
Commercial Draw(Mercantilism)	+1 Gold/turn Migration: +1.5 Gold poolper arrival
Selective Asylum(Piety)	+1 Influence/turn Migration: refugee pull tilt

Modern cards

Policy (Civic)	Effects (native + migration)
Pro-Immigration Stance(Social Question)	+3 Influence/turn Migration: pull x 1.5(openBordersOpenness)
Anti-Immigration Stance(Social Question)	-4 Influence/turn, +4 Production/city Migration: inbound pull x 0.4 (floor 0.15);own cross-civ outbound x 0.6 (retention)
Talent Attraction(Modernization)	+2 Science/turn Migration: +1.5 Science poolper arrival
Cultural Magnetism(Natural History)	+2 Culture/turn Migration: +1.5 Culture poolper arrival
Commercial Draw(Capitalism)	+2 Gold/turn Migration: +1.5 Gold poolper arrival
Refugee Compact(Political Theory)	+2 Influence/turn, +1 Culture/turn Migration: refugee pull tilt

Internal IDs: OPEN_BORDERS = Pro-Immigration Stance, CLOSED_BORDERS = Anti-Immigration Stance, TALENT = Talent Attraction, CULTPULL = Cultural Magnetism, TRADEPULL = Commercial Draw, ASYLUM = Selective Asylum / Refugee Compact. Prefix `TRADITION_EMIG_`; suffix `_ANTIQUITY` / `_EXPLORATION` / `_MODERN`.

How attraction policy card yields function Attraction cards have two yield layers, and they stack:

1. **Native fixed yield from the DB card** (`data/emigration-policies-gameeffects.xml`): a constant per-turn yield in the normal game breakdown (values scale by age).
2. **Carried dividend from immigrant intake** (`emigration-dividend.js`): each incoming migrant under an active attraction adds pool:

$$\mathrm{pool}_y \leftarrow \mathrm{pool}_y + \mathrm{dividendPerMigrant}$$

Each turn the pool decays and grants capped yield:

$$\mathrm{pool}_y \leftarrow \mathrm{pool}_y \cdot \mathrm{dividendDecay}^{\Delta t},$$
$$\mathrm{grant}_y = \min(\mathrm{dividendCap},\ \mathrm{pool}_y)$$

So the card converts migration throughput into ongoing yield. Defaults: `dividendPerMigrant = 1.5`, `dividendDecay = 0.7`, `dividendCap = 12`/turn per channel. The flat `+N Influence/turn` is the card modifier, not per-immigrant; only the carried dividend is per-immigrant.

6c. Environmental disasters & plague (`disastersEnabled`)

Civ VII's `RandomEvents` (flood / volcano / plague / hurricane / blizzard / tornado / duststorm / thunderstorm) become a migration driver parallel to war. `emigration-disasters.js` accumulates per-city disaster distress that decays each turn (game-speed-adjusted) and feeds a situational penalty, so struck cities shed climate/disaster refugees. Fog-independent: the signal is `city.isInfected` plus a severity-scaled spike from `RandomEventOccurred`. Plague-as-contagion (`plagueCarryEnabled`, off): migrants fleeing an infected city seed a smaller outbreak-distress at their destination.

6d. The outlet: crisis death (`attritionEnabled`)

A death channel (cause `attrition`, tracked as deaths, kept out of the migration/refugee metrics) runs on its own `deathPressure` alongside emigration. It fires under lethal distress (`distress >= attritionMinDistress`): war, disaster, siege, famine. Economic emigration carries no situational distress, so it never kills. Two modes:

- **Trapped** (no viable destination): the whole trapped population dies off at full rate.
- **Crisis while fleeing** (a refuge exists, `crisisDeathEnabled`): because the trap almost never fires, a besieged/starving/disaster-struck city loses some people to death while the rest flee, at `crisisDeathShare` (default 0.2) of the trapped rate, so flight dominates. Death builds on `deathPressure`, crosses `attritionThreshold` (×S), and removes a rural point via the same `addRuralPopulation(-1)` the game's starvation uses.

Onset smoothing. Per-turn buildup is scaled by `deathRamp(crisisTenure)`, in [`deathRampFloor` (0.25), 1]. A fresh crisis kills gently and deepens over `deathRampTurns` (6) of sustained distress; `crisisTenure` counts sustained lethal turns and relaxes by one on any turn of relief.

Not capped. Crisis death doesn't count against the war siege-loss cap and isn't held to the emigration rural floor, so a long enough siege or famine can wear rural population all the way down. It removes only rural population, so the urban core and settlement survive until a capture. Lower `crisisDeathShare` or raise `deathRampTurns` if too lethal. State (`deathPressure`, `crisisTenure`) persists across save/reload.

Tuning note: `starvationModifier` is -90 (was -200; -200 flips prosperity negative on its own, so with death on this channel the penalty's job is purely emigration).

6e. Asylum and relationship permeability

Pull is a prosperity gradient plus a targeted-attraction channel (`tilt`) and relationship-permeability multipliers: asylum push (`asylumPushWeight`) for distressed refugees toward hospitable destinations, relationship permeability (`permOpenBorders`, `permAlly`, `permWar`) scaling cross-civ movement, and global bounds (`tiltCap`, `permeFloor`, `permeCeil`). Computed in `emigration-pull.js`, so it composes with prosperity/geography/congestion rather than bypassing them.

6f. Ethnic composition, integration & the per-tile lens (`emigration-composition.js`, `emigration-ethnicity-lens.js`)

Every settlement keeps a running ethnic composition: population by the civilization each person descends from (`emigration-composition.js`), netted each pass from arrivals (origin = source owner), births (current owner), losses (proportional), and conquest (origin buckets kept, owner flips). It follows the settlement, not the owner, so a captured city keeps its residents' origins. The Ethnic Composition lens (Shift+E) paints it as a per-tile mosaic (`emigration-ethnicity-distribution.js`): tiles weighted by build-up (city center > urban > rural > wilderness); each origin claims a share-proportional count of tiles (floored to one) spread across the density gradient, opacity tracking density. Ethnic integration drifts a small fraction of every non-owner origin toward the owner each turn (`integrationRate`), held apart while the host is at war with that origin's homeland (`integrationWarRate`) and slowed in unrest (`integrationUnrestRate`). Toggle: Options ethnic integration (on by default).

6g. Return migration (`emigration-return.js`, `returnEnabled`)

When an origin civ's homeland is at peace with the host and faring well (non-negative net happiness, fed), a fraction of its people abroad set out for home, moving real population (a rural point from the host to one of the homeland's cities), attributed to the returnees' true origin so composition and lens follow them home. Throttled by a per-host cooldown (`returnCooldownTurns`) and a deterministic per-pass rate (`returnRate`), and floored so it only draws from a host that has rural population to give (never invents people, never targets a city-state as a homeland). Toggle: Options return migration (on by default).

6h. Refugee decisions & the Migration Chronicle (`emigration-dilemma.js`, `emigration-chronicle.js`)

The Migration Chronicle (`emigration-chronicle.js`, its own dashboard tab) writes significant movements as short prose (`emigration-narrative.js`): a great exodus, a diaspora taking root, a people returning home. A refugee decision (`emigration-dilemma.js`) is a rare modal triggered by a real upheaval (a neighbor's conquest spree, or a plague crisis) that sends a wave toward the local player: welcome them (a small gold cost, settles a point into your largest city), settle the frontier (a smaller cost, into a town), or turn them away. Hard-capped per age (`dilemmaMaxPerAge`) with a long cooldown (`dilemmaCooldownTurns`), effects light, dismissible (Escape or click outside). For both surfaces, an unmet civ is named as hearsay rather than revealed. Toggle: Options refugee decisions (on by default).

6i. Cultural Enclaves (`emigration-quarter.js`, `emigration-quarter-registry.js`, `emigration-quarter-bonuses.js`)

When a foreign diaspora grows into a lasting, established community in one of your cities (a standing presence, not a lifetime-arrivals total), it forms a Cultural Enclave on a specific edge tile, named for the origin people (e.g. *the Roman Enclave*). You're offered a one-time choice: two identity-grounded options, each a small benefit paired with a matching drawback grounded in that civilization's character (a Roman enclave offers Production/Gold, a Persian one Gold/Culture; 44 civs in `emigration-quarter-bonuses.js`), plus a passive ”let them be.” The chosen stance applies its yields every turn (bounded, ±1–2), so it reads in the city; an unknown/DLC origin falls back to a neutral pair.

Below the prose sits a single short, real, attributed historical quote, shown in the origin people's own language with an English translation (e.g. ” . *(The unexamined life is not worth living.)” — Socrates*). Each original was verified against a primary source, and RTL scripts (Arabic, Persian) are bidi-isolated. An origin has two quotes: its first enclave shows quote A, its second shows quote B.

Bounds:

- **One enclave per host tile.** A different origin overtaking the same tile is a change of hands (the Chronicle notes it, a fresh choice is offered), never a stacked second enclave.
- **At most two enclaves per origin civilization** across your empire (per civilization, not overall). Identity is fixed by CivilizationType when the enclave forms (persisted), so the cap stays correct even if the origin player changes civ across an age.
- **Contested in war.** While you're at war with an enclave's homeland it turns contested: a bounded happiness strain (capped across all your enclaves), framed as wartime suspicion, not disloyalty.
- Throttled with a per-age cap and cooldown, ranked below the refugee decision so two modals never race. Toggle: Options Mods Emigration cultural enclaves (on by default). (Internally the code, config keys, and save data still use ”quarter”; only the player-facing name changed.)

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7. Consequences: the gameplay-write cost layer (`emigration-effects.js`)

Civ VII makes raw population free, so the mod adds the missing feedback via `Players.grantYield(pid, YIELD_X, -amount)` (probe-confirmed to deduct, cross-civ; happiness leg inferred):

- **Assimilation cost (duration-based).** Each migrant adds load to the receiving civ (`assimilationLoadPerMigrant × (1 + assimilationCostPerPop × destPop)`). Load decays each turn (`assimilationDecay`, optionally scaled by `integrationSpeed`) and the civ pays per-turn `assimilationHappiness/assimilationGold` per unit (gold leg optionally scaled by `assimilationEase`). Scoped to migrated population only.
- **Migrant-holding penalty.** Per-turn cost per unsettled `UNIT_MIGRANT` a civ holds.
- **Congestion headwind (Algorithm C).** `congestionPenalty` + `assimLoadFor`; the engine subtracts the headwind from a destination's pull.
- **Carried dividend.** Under attraction contexts, incoming migrants build a decaying per-turn positive pool in the matched yield domain (`emigration-dividend.js`).

All apply to every civ on its own turn. Set any knob to 0 to disable.

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8. Reporting & Demographics integration

When the Demographics mod is installed, Emigration contributes via its companion hook (`globalThis.DemographicsMetricsAPI`, an order-independent handshake):

- **A top-level Emigration tab** (`registerPanel/registerMetricGroup`). Its first section, **Data**, is a metric group with two pill-row toggles: metric and units, **Scaled** (historical people) or **Civ numbers** (raw points). Each metric carries a one-line definition:
  - **Net Migration (Graph):** cumulative arrivals minus departures per civ, over time.
  - **Net Migration (Table):** the same net per civ; the units pills drive the values, magnitude drawn as a diverging bar in its own column (red left of centre for loss, green right for gain).
  - **Emigration:** gross people who left each civ (with a `Sources: War …, Disaster …` breakdown).
  - **Immigration:** gross people who arrived (same breakdown).
  - **Refugees (Left):** people this civ displaced, with war + disaster onset markers (named, by year). War onsets come from Demographics war history; disaster onsets from this mod's event log. Each family has its own toggle (Options Mods Demographics), both default on.
  - **Refugees (Arrived):** displaced people it took in, same toggleable markers.
- **The full dashboard as native sub-tabs:** Network (animated dot-swarm + arrow flow map, each with a Civ Pop / Scaled Pop toggle), Civilizations, Causes, Settlements, Immigration Policies, Notifications, Chronicle, Guide. Registered order-independently; a silent no-op on an older Demographics.

- **A Migration Chronicle** (the Chronicle sub-tab, `emigration-chronicle.js` → `emigration-chronicle-view.js`): a written history of significant movements, distinct from the per-event log. Keeps only the moments that read as history and renders each as prose (`emigration-narrative.js`), spoiler-guarded. Persisted, newest-first, capped.
- **Causes drill down to the event.** Each broad cause on the Causes tab expands to the named events behind it (a particular war, eruption/flood, or the active age crisis), with each event's emigration and deaths. A crisis is attributed to its mechanism (Invasion under War, Plague under Disaster, Loyalty/Revolt under Unhappiness), resolved at the moment of each move. Per-civ event tallies are persisted (`outByEvent` / `deathsByEvent`, capped per civ).
- **A Notifications log** (the Notifications sub-tab): a permanent, scrollable record of every migration notification that has fired. Each row is cause-themed and names the specific event: the named war (via the aggressor map + engine war name) or the named disaster/plague (the game's own `RandomEvents` name). Clicking a row expands it: cause, event, source settlement, destination, and count (both systems). Persisted across save/reload (`emigration-notifications.js` → `-view.js`).
- **An Ethnic Composition lens + plot tooltip** (`emigration-ethnicity-lens.js`, `-tooltip.js`, fed by `emigration-composition.js`). A self-registering lens (Shift+E) paints each settlement as a per-tile mosaic; hovering a settled tile adds exact per-origin percentages to the tooltip. Both honor the spoiler-protection visibility policy (§10).
- **A Refugees row in the Demographics war-effects tooltip** (reads `globalThis.EmigrationData.refugeesCumFor`), rendering "- no data" when Emigration isn't installed.
- **A Network-only timeline-detail note** when the snapshot interval is coarser than every turn (`globalThis.EmigrationTimelineNote`).

`EmigrationData` (global) carries per-civ cumulative tallies: gross in/out, net, refugees, deaths, and the per-cause breakdowns (`emigrationByCauseFor`, `immigrationByCauseFor`). If Demographics isn't installed it's a silent no-op. The dot-swarm (`emigration-network-viz.js`) animates only people who moved: each cross-civ immigrant flies out of its origin civ's circle (its origin sub-cluster when known) to the destination, and each intra-civ mover flies from its source settlement (on load and scrub, not only live playback). Home-grown population materializes in place rather than streaming out of the civ's centre. (Origin lookup uses nullish-coalescing, not `||`, so an origin at node index 0 isn't mistaken for the destination.)

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9. In-game feedback & notifications

Migration is surfaced as styled HUD toasts and, for big world events, world-news. The toast reads as a native Civ VII message: `TitleFont` eyebrow over a `BodyFont` body, the dark panel gradient with bronze/ gold trim (`#8c7e62` frame, `#f0bc78` highlight), slide-in/fade-out, ~11-second dwell, and vertical stacking so several don't overlap. Each toast is themed by cause (a coloured left accent bar + eyebrow: War red, Disaster amber, Attraction green, Conquest, …), and every count is shown in both systems (raw points and scaled people, e.g. *"3 population points (36,000 people)"*). It is important-only by design, with several anti-spam layers, and because toasts stay brief every notification is also recorded in the Notifications log (§8). The layers:

- **Rich, named events** (`emigration-naming.js`): disasters use the game's own names (`GameInfo.RandomEvents.lookup(type).Name`), wars reuse the war name, conquest names the sacked city. E.g. *"The Thera eruption displaces 80,000."*
- **Explanatory, per event** (`emigration-feedback.js`, `emigration-causes.js`). When your cities lose people in a pass, the loss is broken into distinct events (one per source settlement + cause), each with its own accurate count. On screen only the largest event toasts (subject to the cooldown); every event is recorded individually in the log. The cause-keyed action hint and permanence cue ride the toasts and disaster alert.
- **Per-city readout** (`emigration-city-readout.js`). An on-demand panel answering "why is this settlement changing?": the cause mix when more than one pressure is active (*"War 60% · Prosperity 40%"*, else the single dominant cause) + status (building pressure / resting), where its people are pulled, the integration cost, the civ's net migration, the hint, and an at-risk / trapped warning. Built from the recompute-on-read `citySnapshot` (no new state); opens via `emigration.city(id) / .hideCity()` and best-effort on city selection. Toggle `cityReadoutEnabled`; works without Demographics.
- **Dashboard window** (`emigration-window.js` + the shared render core). A standalone HUD window (`emigration.window() / .closeWindow()`) with the whole picture: an animated migration network and cross-civ

flow map (both with a timeline scrubber), a per-civ ledger (in/out/net/refugees/deaths), the cause breakdown, who holds Pro-/Anti-Immigration stances, and cities ranked by pressure. The same core backs the Demographics tab (§8). The timeline records a per-civ population snapshot every pass, including peaceful turns, so the scrubber is available from the opening turns.
- **Anti-spam.** Disasters notify only at/above `disasterNotifyMinSeverity`. World refugee notifications are once-per-milestone on a civ's cumulative refugees (`worldRefugeeThreshold`), but the cumulative total only gates the alert; the headline names the specific war/disaster and reports that pass's outflow, so the figure stays event-scale. A global `notifyCooldownTurns` backstops everything. `notifyMode`: 0 off / 1 important-only (default) / 2 verbose.

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10. Options & tuning

Everything is under **Options → Mods**, in both the main-menu (pregame) and in-game Options screens. `emigration.modinfo` loads the options layer in both shell and game scopes, registering via `Options.addOption({ category: CategoryType.Mods, … })`. Settings persist in the shared, cascade-safe `modSettings` localStorage slice and apply immediately and at game boot.

- **Emigration** group: **Migration counts** (Both / Civ only / Historical only), **Emigration intensity** (Custom / Low / Medium / High), **Dashboard data** (Live / Sample preview), **Migration timeline detail** (snapshot every 1–5 turns), and a **Migration dock button** toggle (on by default).
- **Emigration - Advanced:** every tunable as a dropdown/checkbox, generated from a declarative spec (`emigration-tunables.js`), grouped: pacing, scope, prosperity weights, advanced-model switches (§5), war/violence (incl. siege model + aggressor avoidance), border policies, geography, integration/migrant cost, disasters (+ plague carry), notifications, and the outlet (attrition).

Game-speed scaling is automatic, not a knob: the §2 scaling reads the active speed and applies itself, gated by internal flags (`gameSpeedTuningEnabled` on; `gameSpeedScalePopulation` off) for rollback/QA rather than exposed in the UI.

Simulation scope and visibility are independent. By default the simulation runs over the whole world (every alive civ, from turn one), so topology isn't biased by exploration. (Set *Scope* to met-only to lighten per-turn cost.) Independently, the dashboard and lenses mask civs for spoiler protection per a shared analytics-visibility policy (All / Met-only / Own-civ / Disabled), host-authoritative in multiplayer, default met-only.

The full default set lives in `emigration-config.js`; scaling constants are not exposed (they must match Demographics).

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11. Architecture / module map

Modules are small and single-concern (a 500-line file gate), so several systems span a parent plus helpers. The `ImportFiles` manifest is a complete inventory of the deployed UI tree, gated by a test (`tests/modinfo.mjs`). Key modules:

- `ui/emigration-main.js`: entry UIScript, per-turn hook/costs, event subscriptions, reporting/feedback orchestration, dev dock, boot.
- `ui/emigration-config.js` / `-config-types.js`: tunable defaults + scaling constants, and the `EmigrationConfig` typedef.
- `ui/emigration-game-speed.js`: the game-speed scalar (§2); reads `GameSpeeds.CostMultiplier`, caches S, exposes `speedTurns` / `speedBar` / `speedDecay` / `speedScaleTurn` (fail-safe to 1).
- `ui/emigration-causes.js`: the migration-cause taxonomy (`MigrationCause` + `causeLabel` / `causePermanence` / `causeHint` / `isRefugeeCause`).
- `ui/emigration-tunables.js`: declarative exposed knobs plus Low/Med/High presets.
- `ui/emigration-cities.js`: enumerates met cities into `CitySignal` records.
- `ui/emigration-prosperity.js`: prosperity scoring (legacy + shaped happiness + overcrowding) and `distress`.
- `ui/emigration-violence.js` / `-violence-signals.js`: violence state machine (accumulate/decay, siege tenure/escalation/cap) + the fog-independent polled combat signals.
- `ui/emigration-disasters.js`: per-city disaster distress, decay, plague-carry seeding.
- `ui/emigration-geography.js`: distance decay, flee-from-invader, aggressor preference, Open Borders flow bonus.

- `ui/emigration-civ-tuning.js` / `-war.js`: per-leader/civ tuning table; aggressor map from `DiplomacyDeclareWar`/`MakePeace`.
- `ui/emigration-borders.js`: border-policy reads → `immigrationOpenness` (inbound) + `emigrationRetention` (outbound), attraction yields, asylum flag.
- `ui/emigration-effects.js` / `-dividend.js` / `-migrant-units.js`: assimilation load + congestion headwind; carried-dividend benefit; unsettled-migrant penalty.
- `ui/emigration-engine.js`: main pass, ranking, the two concurrent tracks, per-civ budgets, transit, and the outlet.
- `ui/emigration-arrivals.js`: lagged-arrival processing (the depart/arrive halves of a transit move).
- `ui/emigration-pull.js`: destination decision (`adjustedPull`, `bestDestination`, `migrationCause`).
- `ui/emigration-state.js`: engine-state persistence (per-source voluntary + crisis pressure/cooldown, scaling turn, per-owner populations).
- `ui/emigration-population.js`: population reads/writes and Demographics scaling (with the optional speed-normalized exponent).
- `ui/emigration-migration-stats.js` / `-migration-records.js`: per-civ tallies, the recent-moves feed, the `EmigrationData` global, and the `Migration` record typedef.
- `ui/emigration-city-readout-data.js` / `-city-readout.js`: the per-city snapshot (`buildCitySnapshot` + `citySnapshot`, incl. the cause mix) and the on-demand readout panel.
- `ui/emigration-views.js` / `-ledger-view.js` / `-window.js`: the shared dashboard render core and the standalone window host.
- `ui/emigration-network-viz.js`: the animated dot-swarm; movers fly from their origin civ/settlement on load/scrub; residents materialize in place.
- `ui/emigration-migration-page.js` / `-demographics.js`: the Demographics-page panel registration and the graph specs with subtitles + per-cause tooltips.
- `ui/emigration-prosperity-lens.js` / `-prosperity-tooltip.js`: the tile-by-tile Prosperity lens (§3) and its tooltip.
- `ui/emigration-ethnicity-lens.js` / `-ethnicity-tooltip.js` / `-composition.js`: the Ethnic Composition lens, tooltip, and origin-mix ledger.
- `ui/emigration-naming.js` / `-feedback.js` / `-events.js` / `-report.js` / `-log.js`: rich event naming; cause-themed, dual-number, stacking toasts + world-news + anti-spam; `RandomEventOccurred` handling; record→log lines; dev logging.
- `ui/emigration-notifications.js` / `-notifications-view.js`: the persistent notification log and its click-to-expand sub-tab.
- `ui/emigration-settings.js` / `-options.js` / `ui/options/*`: number-display preference + tunable/preset getters; Options registration; the Advanced editor and the cascade-safe `modSettings` store.
- `data/emigration-policies-*.xml`, `-policies-gameeffects.xml`, `-policy-icons.xml`, `-civilopedia.xml`: DB components for the stance/attraction cards, native modifiers, icons, and Civilopedia pages.
- `devtools/migration-probe.js`: dev-only API probe (separate modinfo, not shipped).
- `text/<locale>/ModText.xml` + `scripts/i18n_*.mjs`: localized strings (all 10 locales) and the dev-only localization pipeline (§14).

`emigration.modinfo` ActionGroups: the options layer in both shell + game scopes; the engine and its submodules (`ImportFiles`) in game scope; an always-on `<UpdateDatabase>` for the Civilopedia pages + policy `TraditionModifiers`; an `<UpdateIcons>` for card art; and three age-scoped `<UpdateDatabase>` groups for the border policies (one per age via `AgeInUse`, required because Civ VII rebuilds the gameplay database each age, so a civic-tree unlock can only reference nodes that exist in that age's database).

The Civilopedia component adds an Emigration section with an overview plus one page per system (Prosperity, War & Refugees, Integration, Borders & Influence, Disasters & Plague, the Outlet, Leaders & Civilizations), reusing the base `Concept` layout. All page text flows through the §14 pipeline (`LOC_PEDIA_EMIG_*`, all 10 locales).

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12. Runtime behavior in game

The mod runs in the UI VM (GameFace JS) and applies migration, costs, and reporting during normal play. Behavior checks are documented in the companion validation notes.

- **`city.addRuralPopulation(±1)`:** the population move/removal. Not owner-gated → works cross-civ. The only population write, which is why the outlet kills via the same channel as starvation.

- **`Players.grantYield(pid, YIELD_X, ±n)`:** yield costs; not owner-gated, negative deducts (gold confirmed, cross-civ).
- **`DiplomacyTreasury.changeDiplomacyBalance(±n)`:** Influence write (superseded; border-policy Influence is now a native `TraditionModifier`).
- **War aggressor:** `DiplomacyDeclareWar` carries `actingPlayer` (declarer) + `reactingPlayer` (target).
- **Game speed:** `Configuration.getGame().gameSpeedType` → `GameInfo.GameSpeeds.lookup(type).CostMultiplier`.
- **Reads (fog-independent):** district health, `city.isInfected`, `Culture.getActiveTraditions` / `isTraditionActive`, `GameInfo.RandomEvents`, `player.leaderType`/`civilizationType`, per-plot yields (`GameplayMap.getYields`), and a per-civ population aggregate, all read for all players.
- **Yields are NET, not gross.** Per-city economic yields are read via `city.Yields.getNetYield(YIELD_X)` (income − maintenance/upkeep), falling back to `getYield` only when net is unavailable. `getYield` alone is gross; an earlier audit found the mod was using it, which made `net food < 0` (starvation) impossible to observe and overstated gold/Prosperity.
- **Happiness economy:** pop upkeep happiness `= 0`, `OVERCROWDING_THRESHOLD = 2` (the grounding for Algorithm B).

What the VM cannot do (and the mod doesn't): create units for other civs (`CREATE_ELEMENT` is local-only), or raise a custom notification type without a DB entry (so engine notifications are deferred; toasts + world-news cover feedback). The policy cards are the one piece that needs the database; everything else is the direct-mutator surface.

---

13. Persistence

Per-game state lives in the `GameConfiguration` KV store (survives save/reload):

- `EmigrationState_v1`: per-source voluntary and crisis pressure/cooldown (§2), plus the monotonic scaling turn.
- `EmigrationViolence_v2`: per-city violence intensity + decay, siege tenure / onset population / cumulative war-loss.
- `EmigrationDisaster_v1`: per-city disaster distress + decay.
- `EmigrationAssim_v1`: per-civ assimilation load + per-civ tick turn.
- `EmigrationDividend_v1`: per-civ carried-dividend pools.
- `EmigrationWar_v1`: victim → aggressors map (§6a).
- `EmigrationEthnos_v1`: per-settlement origin-composition ledger.
- `EmigrationMigStats_v1`: per-civ tallies (net, gross in/out, refugees, deaths, per-cause breakdowns, graph-sample watermarks, and the cumulative city-pair flow matrices, capped at ~4000 edges with lowest-volume evicted).
- `EmigrationNews_v1`: world-news announced-milestone tiers + the last-toast turn (anti-spam).
- `EmigrationNotif_v1`: the permanent notification log (newest-first, capped), each fired notification's cause, turn, summary, count, and origin/destination detail.

Missing fields (e.g. an old save with no crisis-track pressure) are normalized on load, so the two-track split is save-compatible with pre-split games. Options/settings persist separately in the shared `modSettings` localStorage key. Single-player scope (UI-VM gameplay writes are client-side).

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14. Localization

All user-facing strings are LOC keys, fully translated into all 10 locales (en, de, es, fr, it, ja, ko, pt, ru, zh), including the advanced tunable labels. The non-English files are generated, not hand-edited:

```
node scripts/i18n_extract.mjs   # text/en_us/ModText.xml → i18n/i18n-source.json (key list)
node scripts/i18n_apply.mjs     # i18n/<locale>.json → text/<locale>/ModText.xml
```

Author English in `text/en_us/ModText.xml`; translations live in `i18n/<locale>.json` (a missing key falls back to English). `npm run verify` includes a parity gate (`tests/i18n.mjs`) that fails if any en_us key is absent from a locale. `{1_…}` placeholders and code tokens are preserved verbatim.

---

15. Install & run

1. Copy this folder to `~/Library/Application Support/Civilization VII/Mods/emigration/`, relaunch, and enable **Emigration** in *Additional Content*.
2. Play turns. With `const DBG = true` (dev default) the mod logs to `UI.log` via the CSS-parse channel: `grep -E "EMIG_" "~/Library/Application Support/Civilization VII/Logs/UI.log"` `release.sh` flips `DBG` to `false`, so shipped builds run silently.
3. **Dev dock buttons** (subsystem dock): run a pass now / dump the prosperity ranking. Console: `emigration.runNow()`, `emigration.rank()`, `emigration.window()`, `emigration.city(id)`.
4. **Tune or disable layers:** Options → Mods → Emigration - Advanced exposes every advanced-model switch (§5) and interactive system (§6), plus the notification mode. All default on.

Look for `EMIGRATION … left … for …`, `assimilation: …` cost lines, and `ATTRITION … (no refuge)` when the outlet fires.

---

16. Development

Typed JavaScript with JSDoc, no build step: what ships is what you write (see CONTRIBUTING.md). Before committing:

```
npm install
npm run verify      # tsc --noEmit + eslint + the node test harnesses
```

`verify` runs the modularization gate (file/function length / complexity / statements) and 32 test harnesses, including: `game-speed` (the scalar, fail-safe, the 5 speeds, the kill switch, game-time invariance), `notifications` (the persistent log, order, turn-stamp, detail, persistence, ring cap), `network-anim` (immigrants fly from origin), `engine-pass` (a 7-scenario end-to-end pass: peacetime / single-front war / multi-front war / disaster-only / concurrent war + prosperity / famine death + flight / war death + flight), `engine-pull`, `causes`, `city-readout-data`, `city-readout`, `views`, `migration-page`, `scaling` (incl. `formatBoth`), `prosperity`, `geography`, `violence` (siege escalation

- cap), `tunables`, `migration-stats` (incl. the flow-matrix cap), `flow-history`, `composition`, `governance-mask`, `city-scope-global`, `effects`, `civ-tuning`, `war`, `disasters`, `borders`, `naming`, `feedback`, `dividend`, `raid`, `modinfo` (the `ImportFiles` manifest stays a complete, import-closed inventory), `i18n` (locale parity), and `no-empty-catch`. `./release.sh` produces the debug-muted, allow-listed Workshop zip (readable JS, no minification).

The `migration-probe` mod (its own modinfo, never shipped) is the in-engine verifier behind the write-surface/data claims: dock buttons + a `globalThis.mig` console API, with the `API3` and `API4` confirmation passes + passive `DiplomacyDeclareWar` / `RandomEventOccurred` recorders. Audit commands: `mig.warName()` confirms the engine war name resolves (`getJointEvents → uniqueID → getWarData(uniqueID, me).warName`) and dumps `getWarData`; `mig.happy()` grants `YIELD_HAPPINESS` and re-reads to confirm whether happiness is a grantable stockpile; `mig.blob()` logs every persisted value's size + flow-key counts to confirm saves stay bounded; the passive `API4-B DeclareWar` dump logs the real war-event payload keys.

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17. Caveats & limits

- **Single-player.** UI-VM gameplay writes are client-side.
- **The advanced layers are on by default but un-playtested-at-scale.** Everything beyond the baseline is implemented and unit-tested, but the knob values are starting points. Turn pieces off in Options (§5–§6) if a save hits balance trouble.
- **Game-speed scaling is verified by tests, not yet in-game at every speed.** The scalar + transforms + game-time invariance are unit-tested and fail-safe to $S = 1$; `gameSpeedTuningEnabled` is the kill switch if a speed feels off.
- **Happiness cost is inferred (and probe-checkable).** Negative gold grants are probe-confirmed; negative happiness is inferred: the base game exposes no player-happiness mutator, so `grantYield(YIELD_HAPPINESS, -n)` may be a no-op. Run `mig.happy()` to confirm; if nothing moves, set the happiness knobs to 0 (the congestion headwind, §5-C, is the gold-immune structural brake).

- **War names use the engine's when resolvable.** Resolves via `getJointEvents → DECLARE_WAR uniqueID → getWarData(uniqueID, localPlayerID).warName`, falling back to "{Victim}–{Aggressor} War". The aggressor map reads `actingPlayer/reactingPlayer` with `initialPlayer/targetPlayer` fallbacks.
- **A few in-engine confirmations remain best-effort:** the policy cards' in-game slotting/unlock, whether disaster plot-effects are pollable per plot (we use `isInfected`, confirmed), the per-plot yield shape for the Prosperity lens (`getYields`, used defensively with a per-city fallback), and a longitudinal getYield pre/post-penalty check. The code degrades to a safe no-op where unconfirmed.
- **On-map floating indicators are deferred:** `WorldUI` exposes no floating-text method, so feedback uses toasts.
- **Engine notifications need a DB type:** clickable end-turn notifications would need a `NotificationType`; toasts + world-news cover it for now.
- **No new game rules.** The mod composes existing engine writes; it can't invent effects or spawn units for the AI.

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18. Compatibility & mod coexistence

Its only dependency is `base-standard`, and it touches every shared surface additively:

- **Database: inserts only.** `data/emigration-policies-*.xml` and `data/emigration-civilopedia.xml` are pure `<Row>` inserts: no `<Replace>`, `<Update>`, or `<Delete>` against base or shared tables. New IDs are namespaced (`TRADITION_EMIG_*`, `SectionID="EMIGRATION"`). Border-policy traditions attach to existing civic-tree nodes via additive `ProgressionTreeNodeUnlocks` rows.
- **Shared settings store, self-healing.** Options persist under the single community-convention `modSettings` localStorage key (§13), never a stray top-level key. GameFace's `localStorage` is shared across every mod, so a stray top-level key would clobber the store; on each save the store self-heals, preserving `modSettings` and dropping stray keys (ui/options/mod-options.js). With any mod that follows the convention (Demographics does) this is a no-op.
- **Cooperative globals + events.** JS globals are namespaced (`globalThis.emigration`, `EmigrationData`). The Demographics integration uses an order-independent handshake (`globalThis.DemographicsMetricsAPI ??= {}`) that joins the metrics API rather than replacing it. Engine events (`engine.on(…)`) are multicast, so subscribing never blocks another mod's handlers.
- **Defers to the Demographics namespace.** The war-popup refugees label and glossary are owned by Demographics; Emigration adds only the one graph-title string it introduces, so there are no duplicate `LocalizedText` definitions.
- **Additive plot tooltip.** Both the Ethnic Composition and Prosperity per-tile breakdowns are appended into the live plot tooltip's `.tooltip__content` via a `MutationObserver`, never replacing the tooltip, so Emigration stacks with whichever full-tooltip mod is active (bz-map-trix, TCS Improved Plot Tooltip).
- **Adaptive to other mods.** It reads live happiness / yields / Influence each turn, so a mod that rebalances those values is reflected in the Prosperity score; effects compose rather than conflict. If another mod also moves population or changes yields, the two stack without corrupting state. The §2 scaling reads whatever game speed is active, including custom speeds a mod adds (any `GameSpeed` row with a `CostMultiplier`).

Source

Open source on GitHub: https://github.com/tmtmiller1/civilizationvii-emigration

Credits

- Tower, for design and Civilization VII implementation.
- Tomahawk, Mk Z, and Tim_The_Texan, creators of the Civilization V Emigration mod that inspired this project.

Special Thanks

- **Potato McWhisky**, for teaching me to love again, Civilization-wise (Civ VI), after growing up as a Civilization II, IV, and V player. Making this mod is an act of faith that the community will eventually help make Civilization VII as good as the previous entries.

License

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